Inverse Trigonometric Functions: Questions and Shortened Solutions

Question 01

English

Let $\alpha = -\dfrac{1+\sqrt{5}}{4}$. Considering only the principal values of the inverse trigonometric functions, the value of $\sin^{-1}\alpha + \cos^{-1}\left(2\alpha^2 - 1\right)$ is

(A) $-\dfrac{\pi}{5}$
(B) $\dfrac{\pi}{5}$
(C) $\dfrac{\pi}{10}$
(D) $\dfrac{13\pi}{10}$

Only one option is correct.

हिन्दी

माना कि $\alpha = -\dfrac{1+\sqrt{5}}{4}$ है। प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मानों को लेते हुए, $\sin^{-1}\alpha + \cos^{-1}\left(2\alpha^2 - 1\right)$ का मान है

(A) $-\dfrac{\pi}{5}$
(B) $\dfrac{\pi}{5}$
(C) $\dfrac{\pi}{10}$
(D) $\dfrac{13\pi}{10}$

केवल एक विकल्प सही है।

Solution

$-\alpha = \dfrac{1+\sqrt{5}}{4} = \cos 36^\circ = \sin 54^\circ$, and $54^\circ \in \left[0, \dfrac{\pi}{2}\right]$, so $\sin^{-1}\alpha = -\dfrac{3\pi}{10}$

$2\alpha^2 - 1 = 2\cos^2 36^\circ - 1 = \cos 72^\circ = \sin 18^\circ = \dfrac{\sqrt{5}-1}{4} > 0$

$72^\circ \in [0, \pi]$, so $\cos^{-1}\left(2\alpha^2 - 1\right) = \dfrac{2\pi}{5}$ (not $2\cos^{-1}\alpha = \dfrac{8\pi}{5}$, which is outside $[0, \pi]$)

Sum $= -\dfrac{3\pi}{10} + \dfrac{4\pi}{10} = \dfrac{\pi}{10}$

Correct Option: (C)

Question 02

English

For each positive integer n, let $S_n = \sum_{k=1}^{n} \tan^{-1}\left(\frac{4k}{k^4 - k^2 + 4}\right)$, where $\tan^{-1}$ takes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Then $S_2 + S_3$ is equal to

(A) $\pi - \tan^{-1}\left(\frac{9}{17}\right)$

(B) $\pi - \tan^{-1}\left(\frac{17}{9}\right)$

(C) $\frac{3\pi}{4}$

(D) $\pi - \tan^{-1}\left(\frac{9}{19}\right)$

Only one option is correct.

हिन्दी

प्रत्येक धनात्मक पूर्णांक n के लिए, माना कि $S_n = \sum_{k=1}^{n} \tan^{-1}\left(\frac{4k}{k^4 - k^2 + 4}\right)$ है, जहाँ $\tan^{-1}$ के मान $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में हैं। तब $S_2 + S_3$ बराबर है

(A) $\pi - \tan^{-1}\left(\frac{9}{17}\right)$

(B) $\pi - \tan^{-1}\left(\frac{17}{9}\right)$

(C) $\frac{3\pi}{4}$

(D) $\pi - \tan^{-1}\left(\frac{9}{19}\right)$

केवल एक विकल्प सही है।

Solution

Let $T_k = \frac{k(k+1)}{2}$. Then $T_k - T_{k-1} = k$ and $1 + T_k T_{k-1} = 1 + \frac{k^2(k^2-1)}{4} = \frac{k^4 - k^2 + 4}{4}$

So $\frac{4k}{k^4 - k^2 + 4} = \frac{T_k - T_{k-1}}{1 + T_k T_{k-1}}$ with $T_k T_{k-1} \geq 0$, hence each term $= \tan^{-1} T_k - \tan^{-1} T_{k-1}$

Telescoping with $T_0 = 0$: $S_n = \tan^{-1} \frac{n(n+1)}{2}$, so $S_2 = \tan^{-1} 3$, $S_3 = \tan^{-1} 6$

$3 \cdot 6 = 18 > 1$ with both positive, so $\tan^{-1} 3 + \tan^{-1} 6 = \pi + \tan^{-1} \frac{3+6}{1-18} = \pi - \tan^{-1} \frac{9}{17}$

Correct Option: (A)

Question 03

English

Let S denote the set of all real numbers x satisfying $2\sin^{-1}(\sin x) = \cos^{-1}\left(\frac{x}{20}\right)$, where the inverse trigonometric functions take their principal values. Then the number of elements in S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी वास्तविक संख्याओं x का समुच्चय है जो $2\sin^{-1}(\sin x) = \cos^{-1}\left(\frac{x}{20}\right)$ को संतुष्ट करती हैं, जहाँ प्रतिलोम त्रिकोणमितीय फलन अपने मुख्य मान लेते हैं। तब S में अवयवों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Domain: $\left|\frac{x}{20}\right| \le 1 \Rightarrow x \in [-20, 20]$. Range: RHS $\in [0, \pi]$, so $f(x) = \sin^{-1}(\sin x)$ must equal $g(x) = \frac{1}{2}\cos^{-1}\frac{x}{20} \in \left[0, \frac{\pi}{2}\right]$

$f \ge 0$ only on humps $[2k\pi, (2k+1)\pi]$, where f rises from 0 to $\frac{\pi}{2}$ (at the midpoint) and falls back to 0; g decreases from $\frac{\pi}{2}$ to 0 with $0 < g < \frac{\pi}{2}$ on $(-20, 20)$

Full humps inside $[-20, 20]$: $k = -3, -2, -1, 0, 1, 2$ (from $[-6\pi, -5\pi]$ to $[4\pi, 5\pi]$; $-6\pi \approx -18.85$)

On each: $f - g < 0$ at both ends and > 0 at the midpoint; $f - g$ is strictly monotonic on each half ($|g'| = \frac{1}{2\sqrt{400 - x^2}} < 1$ there), so 2 roots per hump $\Rightarrow 12$

Partial hump $[6\pi, 20]$: $f = x - 6\pi$ rises $0 \to 1.15$, g falls $0.17 \to 0$: 1 root. On $[-20, -6\pi]$, $f < 0$: none

Total $= 12 + 1 = 13$

Final Answer: 13

Question 04

English

Let D be the set of all real numbers x for which $\sin^{-1}\left(\frac{[x]}{2}\right) - \cos^{-1}(x - [x])$ is defined, where $[t]$ denotes the greatest integer less than or equal to t and the inverse trigonometric functions take their principal values. Let $f : D \to \mathbb{R}$ be defined by $f(x) = \sin^{-1}\left(\frac{[x]}{2}\right) - \cos^{-1}(x - [x])$. If the range of f is the interval $[\alpha, \beta)$, then the value of $\frac{\beta}{\alpha}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि D उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $\sin^{-1}\left(\frac{[x]}{2}\right) - \cos^{-1}(x - [x])$ परिभाषित है, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है तथा प्रतिलोम त्रिकोणमितीय फलन अपने मुख्य मान लेते हैं। माना कि $f : D \to \mathbb{R}$, $f(x) = \sin^{-1}\left(\frac{[x]}{2}\right) - \cos^{-1}(x - [x])$ द्वारा परिभाषित है। यदि f का परिसर अन्तराल $[\alpha, \beta)$ है, तो $\frac{\beta}{\alpha}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$x - [x] = \{x\} \in [0, 1)$ is always valid for $\cos^{-1}$; $\left|\frac{[x]}{2}\right| \leq 1 \Rightarrow [x] \in \{-2, -1, 0, 1, 2\} \Rightarrow D = [-2, 3)$

For $[x] = n$, $\{x\}$ takes every value in $[0, 1)$, so $\cos^{-1}\{x\}$ takes every value in $\left(0, \frac{\pi}{2}\right]$

Hence on $[n, n+1)$ the values of f fill $\left[s_n - \frac{\pi}{2}, s_n\right)$ with $s_n = \sin^{-1}\frac{n}{2} = -\frac{\pi}{2}, -\frac{\pi}{6}, 0, \frac{\pi}{6}, \frac{\pi}{2}$

Pieces: $\left[-\pi, -\frac{\pi}{2}\right), \left[-\frac{2\pi}{3}, -\frac{\pi}{6}\right), \left[-\frac{\pi}{2}, 0\right), \left[-\frac{\pi}{3}, \frac{\pi}{6}\right), \left[0, \frac{\pi}{2}\right)$; consecutive pieces overlap

Range $= \left[-\pi, \frac{\pi}{2}\right)$ ($-\pi$ attained at $x = -2$; $\frac{\pi}{2}$ not attained) $\Rightarrow \frac{\beta}{\alpha} = \frac{\pi/2}{-\pi} = -0.5$

Final Answer: -0.50

Question 05

English

Let D be the set of all real numbers x for which both $\sin^{-1}\left(\frac{x\left(3-x^2\right)}{2}\right)$ and $\sin^{-1}\left(\frac{x}{2}\right)$ are defined, where $\sin^{-1}$ takes values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Let $F : D \to \mathbb{R}$ be defined by $F(x) = \sin^{-1}\left(\frac{x\left(3-x^2\right)}{2}\right) - 3\sin^{-1}\left(\frac{x}{2}\right)$. If m is the minimum value of F on D, then the value of $\frac{m}{\pi}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि D उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $\sin^{-1}\left(\frac{x\left(3-x^2\right)}{2}\right)$ तथा $\sin^{-1}\left(\frac{x}{2}\right)$ दोनों परिभाषित हैं, जहाँ $\sin^{-1}$ के मान $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ में हैं। माना कि $F : D \to \mathbb{R}$, $F(x) = \sin^{-1}\left(\frac{x\left(3-x^2\right)}{2}\right) - 3\sin^{-1}\left(\frac{x}{2}\right)$ द्वारा परिभाषित है। यदि D पर F का न्यूनतम मान m है, तो $\frac{m}{\pi}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\frac{x(3-x^2)}{2} \le 1 \iff x^3 - 3x + 2 \ge 0 \iff (x-1)^2(x+2) \ge 0 \iff x \ge -2$

$\frac{x(3-x^2)}{2} \ge -1 \iff x^3 - 3x - 2 \le 0 \iff (x+1)^2(x-2) \le 0 \iff x \le 2$; with $\left|\frac{x}{2}\right| \le 1$: $D = [-2, 2]$

Put $x = 2\sin t$, $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$: $\frac{x(3-x^2)}{2} = 3\sin t - 4\sin^3 t = \sin 3t$, so $F = \sin^{-1}(\sin 3t) - 3t$

$|x| \le 1$ ($|3t| \le \frac{\pi}{2}$): $F = 0$; $1 \le x \le 2$: $F = \pi - 6t$, decreasing to $F(2) = \pi - 3\pi = -2\pi$; $-2 \le x \le -1$: $F = -\pi - 6t \ge 0$

Check: $F(2) = \sin^{-1}(-1) - 3\sin^{-1} 1 = -\frac{\pi}{2} - \frac{3\pi}{2} = -2\pi$. So $m = -2\pi$, $\frac{m}{\pi} = -2$

Final Answer: -2

Question 06

English

Let $S = \sum_{k=1}^{\infty} \cot^{-1}\left(\frac{9k^2 - 15k + 5}{3}\right)$, where $\cot^{-1}$ takes values in $(0, \pi)$. Then the value of $\frac{2S}{\pi}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $S = \sum_{k=1}^{\infty} \cot^{-1}\left(\frac{9k^2 - 15k + 5}{3}\right)$ है, जहाँ $\cot^{-1}$ के मान $(0, \pi)$ में हैं। तब $\frac{2S}{\pi}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $u_k = 3k - 4$: $u_{k+1} - u_k = 3$, $1 + u_k u_{k+1} = 1 + (3k - 4)(3k - 1) = 9k^2 - 15k + 5$

For $b > a$, $\theta = \tan^{-1} b - \tan^{-1} a \in (0, \pi)$ and $\cot\theta = \frac{1 + ab}{b - a}$; since $\cot^{-1}$ has range $(0, \pi)$, $\cot^{-1}\frac{1 + ab}{b - a} = \tan^{-1} b - \tan^{-1} a$ always

So the k-th term $= \tan^{-1}(3k - 1) - \tan^{-1}(3k - 4)$, including $k = 1$ where the argument is $-\frac{1}{3}$

Partial sum $= \tan^{-1}(3n - 1) - \tan^{-1}(-1) \to \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$

$\frac{2S}{\pi} = \frac{3}{2} = 1.5$ (writing $\cot^{-1} x = \tan^{-1}\frac{1}{x}$ at $k = 1$ would wrongly give -0.5)

Final Answer: 1.50

Question 07

English

Let $S = \left\{x \in [0, 2\pi] : x \neq \frac{\pi}{2},\ x \neq \frac{3\pi}{2} \text{ and } \cos^{-1}(\cos x) < 3\tan^{-1}(\tan x)\right\}$, where the inverse trigonometric functions take their principal values. If the sum of the lengths of the intervals that constitute S is $k\pi$, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $S = \left\{x \in [0, 2\pi] : x \neq \frac{\pi}{2},\ x \neq \frac{3\pi}{2},\ \cos^{-1}(\cos x) < 3\tan^{-1}(\tan x)\right\}$ है, जहाँ प्रतिलोम त्रिकोणमितीय फलन अपने मुख्य मान लेते हैं। यदि S को बनाने वाले अन्तरालों की लम्बाइयों का योग $k\pi$ है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$C(x) = \cos^{-1}(\cos x)$: x on $[0, \pi]$, $2\pi - x$ on $[\pi, 2\pi]$; $T(x) = \tan^{-1}(\tan x)$: x, $x - \pi$, $x - 2\pi$ on $\left[0, \frac{\pi}{2}\right)$, $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$, $\left(\frac{3\pi}{2}, 2\pi\right]$

$\left[0, \frac{\pi}{2}\right)$: $x < 3x \iff x > 0$: length $\frac{\pi}{2}$

$\left(\frac{\pi}{2}, \pi\right]$: $T \leq 0 \leq C$: none

$\left(\pi, \frac{3\pi}{2}\right)$: $2\pi - x < 3x - 3\pi \iff x > \frac{5\pi}{4}$: length $\frac{\pi}{4}$

$\left(\frac{3\pi}{2}, 2\pi\right]$: $T \leq 0 \leq C$: none. Total $\frac{3\pi}{4} \Rightarrow k = 0.75$

Final Answer: 0.75

Question 08

English

Let $S = \sum_{k=1}^{7} \cos^{-1}\left(\sin\frac{2k\pi}{9}\right)$, where $\cos^{-1}$ takes values in $[0,\pi]$. Then the value of $\frac{180S}{\pi}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $S = \sum_{k=1}^{7} \cos^{-1}\left(\sin\frac{2k\pi}{9}\right)$ है, जहाँ $\cos^{-1}$ के मान $[0,\pi]$ में हैं। तब $\frac{180S}{\pi}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\cos^{-1} y = \frac{\pi}{2} - \sin^{-1} y$, so $S = \frac{7\pi}{2} - \sum_{k=1}^{7} \sin^{-1}\left(\sin\frac{2k\pi}{9}\right)$

In degrees the angles are $40k^\circ$: $40, 80, 120, 160, 200, 240, 280$

$\sin^{-1}(\sin\theta)$ in degrees: $40, 80, 180-120 = 60, 180-160 = 20, 180-200 = -20, 180-240 = -60, 280-360 = -80$; sum $= 40$

$\frac{180S}{\pi} = 7 \cdot 90 - 40 = 590$

Final Answer: 590

Question 09

English

Considering only the principal values of the inverse trigonometric functions, with $\cot^{-1}$ taking values in $(0,\pi)$, let S be the set of all integers n with $1 \le |n| \le 10$ such that $\cot^{-1}(\cot n) = \tan^{-1}(\tan n)$. Then the sum of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मानों को लेते हुए, जहाँ $\cot^{-1}$ के मान $(0,\pi)$ में हैं, माना कि S उन सभी पूर्णांकों n का समुच्चय है जिनके लिए $1 \le |n| \le 10$ तथा $\cot^{-1}(\cot n) = \tan^{-1}(\tan n)$ है। तब S के सभी अवयवों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Both sides differ from n by a multiple of π; $\cot^{-1}(\cot n) = r \in (0,\pi)$ where $r = n - \pi\left[\frac{n}{\pi}\right]$

$\tan^{-1}(\tan n) = r$ if $r < \frac{\pi}{2}$ and $r - \pi$ if $r > \frac{\pi}{2}$, so equality $\iff r \in \left(0, \frac{\pi}{2}\right)$

$n > 0$: $r = 1, 2, 3, 4-\pi, 5-\pi, 6-\pi, 7-2\pi, 8-2\pi, 9-2\pi, 10-3\pi \approx 1, 2, 3, 0.86, 1.86, 2.86, 0.72, 1.72, 2.72, 0.58$: $n = 1, 4, 7, 10$

$n < 0$: $r = n + m\pi$: $n = -1, \ldots, -10$ give $r \approx 2.14, 1.14, 0.14, 2.28, 1.28, 0.28, 2.42, 1.42, 0.42, 2.57$: $n = -2, -3, -5, -6, -8, -9$

Sum $= (1+4+7+10) - (2+3+5+6+8+9) = 22 - 33 = -11$

Final Answer: -11

Question 10

English

Let $f : [-1, 1] \to \mathbb{R}$ be defined by $f(x) = \cos^{-1}\left(\frac{x + \sqrt{3 - 3x^2}}{2}\right) + \cos^{-1}\left(\frac{x - \sqrt{3 - 3x^2}}{2}\right)$, where $\cos^{-1}$ takes values in $[0, \pi]$. Then the value of $\frac{1}{\pi}\sum_{k=0}^{18} f\left(\cos\frac{k\pi}{18}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [-1, 1] \to \mathbb{R}$, $f(x) = \cos^{-1}\left(\frac{x + \sqrt{3 - 3x^2}}{2}\right) + \cos^{-1}\left(\frac{x - \sqrt{3 - 3x^2}}{2}\right)$ द्वारा परिभाषित है, जहाँ $\cos^{-1}$ के मान $[0, \pi]$ में हैं। तब $\frac{1}{\pi}\sum_{k=0}^{18} f\left(\cos\frac{k\pi}{18}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $x = \cos\theta$, $\theta \in [0, \pi]$, $\sqrt{3 - 3x^2} = \sqrt{3}\sin\theta$: arguments are $\cos\left(\theta - \frac{\pi}{3}\right)$ and $\cos\left(\theta + \frac{\pi}{3}\right)$

$\cos^{-1}\cos\left(\theta - \frac{\pi}{3}\right) = \left|\theta - \frac{\pi}{3}\right|$; $\cos^{-1}\cos\left(\theta + \frac{\pi}{3}\right) = \theta + \frac{\pi}{3}$ for $\theta \le \frac{2\pi}{3}$, $\frac{5\pi}{3} - \theta$ for $\theta \ge \frac{2\pi}{3}$

So $f = \frac{2\pi}{3}$ for $\theta \le \frac{\pi}{3}$, $f = 2\theta$ for $\frac{\pi}{3} \le \theta \le \frac{2\pi}{3}$, $f = \frac{4\pi}{3}$ for $\theta \ge \frac{2\pi}{3}$

$\theta = \frac{k\pi}{18}$: $k = 0, \ldots, 6$ give $7 \cdot \frac{2\pi}{3}$; $k = 7, \ldots, 11$ give $\frac{2\pi}{18}(7 + 8 + 9 + 10 + 11) = 5\pi$; $k = 12, \ldots, 18$ give $7 \cdot \frac{4\pi}{3}$

Total $= \frac{14\pi}{3} + 5\pi + \frac{28\pi}{3} = 19\pi$, so the value is 19

Final Answer: 19

Question 11

English

Let $u : (-1, 1) \to \mathbb{R}$ and $v : (-1, 1) \to \mathbb{R}$ be defined by $u(x) = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ and $v(x) = \tan^{-1}\sqrt{\dfrac{1-x}{1+x}}$, where the inverse trigonometric functions take their principal values. There is exactly one $x_0 \in (-1, 1)$ such that $u(x_0) = v(x_0)$. Then the value of the derivative of v with respect to u at $x = x_0$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $u : (-1, 1) \to \mathbb{R}$ तथा $v : (-1, 1) \to \mathbb{R}$, $u(x) = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ तथा $v(x) = \tan^{-1}\sqrt{\dfrac{1-x}{1+x}}$ द्वारा परिभाषित हैं, जहाँ प्रतिलोम त्रिकोणमितीय फलन अपने मुख्य मान लेते हैं। ठीक एक $x_0 \in (-1, 1)$ ऐसा है कि $u(x_0) = v(x_0)$ है। तब $x = x_0$ पर u के सापेक्ष v के अवकलज का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $x = \cos\phi$, $\phi \in (0, \pi)$: $\sqrt{\dfrac{1-x}{1+x}} = \tan\dfrac{\phi}{2}$ so $v = \dfrac{\phi}{2}$; $2x\sqrt{1-x^2} = \sin 2\phi$

$u = 2\phi$ for $\phi \le \dfrac{\pi}{4}$, $u = \pi - 2\phi$ for $\dfrac{\pi}{4} \le \phi \le \dfrac{3\pi}{4}$, $u = 2\phi - 2\pi < 0$ for $\phi > \dfrac{3\pi}{4}$

$u = v$: $2\phi = \dfrac{\phi}{2}$ only at $\phi = 0$ (excluded); $\pi - 2\phi = \dfrac{\phi}{2} \Rightarrow \phi = \dfrac{2\pi}{5} \in \left(\dfrac{\pi}{4}, \dfrac{3\pi}{4}\right)$; last piece has $u < 0 < v$. So $x_0 = \cos 72^\circ$

Near x_0: $\dfrac{dv}{d\phi} = \dfrac{1}{2}$, $\dfrac{du}{d\phi} = -2$, so $\dfrac{dv}{du} = \dfrac{1/2}{-2} = -0.25$

Final Answer: -0.25

Question 12

English

Let $x_0 = 0$ and, for every integer $n \geq 0$, let $x_{n+1} = \dfrac{4x_n + 3\sqrt{1 - x_n^2}}{5}$. Then the value of x_{100} is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $x_0 = 0$ है तथा प्रत्येक पूर्णांक $n \geq 0$ के लिए $x_{n+1} = \dfrac{4x_n + 3\sqrt{1 - x_n^2}}{5}$ है। तब x_{100} का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $\varphi = \sin^{-1}\dfrac{3}{5}$ $(\cos\varphi = \dfrac{4}{5})$ and $\theta_n = \sin^{-1} x_n$; then $x_{n+1} = \sin\theta_n\cos\varphi + \cos\theta_n\sin\varphi = \sin(\theta_n + \varphi)$

So $\theta_{n+1} = \sin^{-1} x_n + \sin^{-1}\dfrac{3}{5}$ if this is $\leq \dfrac{\pi}{2}$, and $\pi - (\theta_n + \varphi)$ otherwise

$\theta_1 = \varphi$, $\theta_2 = 2\varphi$ $(2\varphi < \dfrac{\pi}{2}$ as $\tan\varphi = \dfrac{3}{4} < 1)$; $3\varphi > \dfrac{\pi}{2}$ so $\theta_3 = \pi - 3\varphi$

$\theta_3 + \varphi = \pi - 2\varphi > \dfrac{\pi}{2} \Rightarrow \theta_4 = 2\varphi$: period 2, $\theta_n = 2\varphi$ for even $n \geq 2$

$x_{100} = \sin 2\varphi = 2 \cdot \dfrac{3}{5} \cdot \dfrac{4}{5} = \dfrac{24}{25} = 0.96$

Final Answer: 0.96

Question 13

English

Considering only the principal value of $\tan^{-1}$, the number of ordered pairs (m, n) of positive integers with $m \leq 50$ and $n \leq 50$ satisfying $\tan^{-1}\dfrac{1}{m} + \tan^{-1}\dfrac{1}{n} \geq \tan^{-1}\dfrac{1}{2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\tan^{-1}$ के केवल मुख्य मान को लेते हुए, धनात्मक पूर्णांकों के उन क्रमित युग्मों (m, n) की संख्या, जिनके लिए $m \leq 50$, $n \leq 50$ तथा $\tan^{-1}\dfrac{1}{m} + \tan^{-1}\dfrac{1}{n} \geq \tan^{-1}\dfrac{1}{2}$ है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

If $m \leq 2$ (or $n \leq 2$): $\tan^{-1}\dfrac{1}{m} \geq \tan^{-1}\dfrac{1}{2}$ and the other term is positive, so the inequality holds: $50^2 - 48^2 = 196$ pairs

If $m, n \geq 3$: $\dfrac{1}{mn} < 1$, so the sum is $\tan^{-1}\dfrac{m+n}{mn-1} \in \left(0, \dfrac{\pi}{2}\right)$ and $\tan^{-1}$ is increasing

$\dfrac{m+n}{mn-1} \geq \dfrac{1}{2} \iff mn - 2m - 2n - 1 \leq 0 \iff (m-2)(n-2) \leq 5$

Equality case $(m-2)(n-2) = 5$: $(3, 7), (7, 3)$, i.e. $\tan^{-1}\dfrac{1}{3} + \tan^{-1}\dfrac{1}{7} = \tan^{-1}\dfrac{1}{2}$ (included by $\geq$)

$a = m-2,\ b = n-2 \geq 1,\ ab \leq 5$: $\displaystyle\sum_{a=1}^{5}\left[\frac{5}{a}\right] = 5+2+1+1+1 = 10$. Total $= 196+10 = 206$

Final Answer: 206

Question 14

English

Let S be the set of all real numbers x satisfying $\sin\left(2\tan^{-1}x\right) = \cos\left(2\tan^{-1}\dfrac{2x}{3}\right)$, where $\tan^{-1}$ takes values in $\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$. Then the sum of the squares of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी वास्तविक संख्याओं x का समुच्चय है जो $\sin\left(2\tan^{-1}x\right) = \cos\left(2\tan^{-1}\dfrac{2x}{3}\right)$ को संतुष्ट करती हैं, जहाँ $\tan^{-1}$ के मान $\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$ में हैं। तब S के सभी अवयवों के वर्गों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $\alpha = \tan^{-1}x$, $\beta = \tan^{-1}\dfrac{2x}{3}$, both in $\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$ with the same sign as x

$\cos\left(\dfrac{\pi}{2}-2\alpha\right) = \cos 2\beta \Rightarrow \dfrac{\pi}{2}-2\alpha = 2k\pi \pm 2\beta \Rightarrow \alpha \pm \beta = \dfrac{\pi}{4} - k\pi$

$\alpha+\beta \in (-\pi,\pi)$: $\alpha+\beta = \dfrac{\pi}{4}$ or $-\dfrac{3\pi}{4}$, both give $\tan(\alpha+\beta) = 1$: $\dfrac{x+\frac{2x}{3}}{1-\frac{2x^2}{3}} = 1 \Rightarrow 2x^2+5x-3 = 0 \Rightarrow$

$x = \dfrac{1}{2}$ or -3

Check: $x = \dfrac{1}{2}$: $\alpha+\beta = \tan^{-1}\dfrac{1}{2}+\tan^{-1}\dfrac{1}{3} = \dfrac{\pi}{4}$; $x=-3$: $\alpha+\beta = -(\tan^{-1}3+\tan^{-1}2) = -\dfrac{3\pi}{4}$. Both valid

$\alpha-\beta$ has $|\alpha-\beta| < \dfrac{\pi}{2}$, so only $\alpha-\beta = \dfrac{\pi}{4}$: $\dfrac{x/3}{1+2x^2/3} = 1 \Rightarrow 2x^2-x+3 = 0$, no real root

$S = \left\{\dfrac{1}{2},-3\right\}$, sum of squares $= \dfrac{1}{4}+9 = 9.25$

Final Answer: 9.25